Ocean Sound Separation with Causal High-Resolution Background Priors

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***Abstract*—We test whether past ocean soundscape can predict a background prior that improves separation of a future event. DCLDE examples combine an annotation-clean real 5-s ocean recording with a synthetic marine event; the preceding 60-s history masks annotated calls. An initial 30-band GRU forecast injected through FiLM does not improve BS-RoFormer on 4,096 held-out mixtures (9.497 versus 9.631 dB target SI-SDRi), even with the true current 30-band spectrum as an Oracle (9.535 dB). An information ladder instead identifies a useful 128-group, time-resolved representation. We feed its background map and explicit mixture-minus-background innovation through a matched residual adapter. The true-background Oracle reaches 16.767 dB; fixed-mean and wrong-day map controls reach 12.156 and 11.944 dB, while past-60-s persistence and a frozen GRU forecast reach 14.295 and 14.351 dB. Persistence exceeds the fixed-mean and wrong-day controls by 2.138 and 2.351 dB, respectively. Forecast exceeds persistence by 0.056 dB (recording-day paired 95% CI $[0.036, 0.074]$), but remains 2.416 dB below Oracle. Thus the residual interface itself helps, and time-relevant historical background supplies a further substantial gain; this GRU adds little beyond persistence. Synthetic targets and only three held-out days limit generalization; classifier tests on untouched calls do not establish real-event superiority. On the same 103 real calls, causal conditions preserve off-event audio better than wrong-day Shuffled, whereas Shuffled scores better on the classifier's two event-probability measures; the real-event ranking is therefore unresolved.**

***Index Terms*—ocean acoustics, source separation, soundscape forecasting, BS-RoFormer, weak-reference evaluation**

I. Introduction

Ocean backgrounds vary with shipping, weather, biology, and season [1]. A mixture-only separator must infer both an unexpected event and this changing background from the same observation. We instead ask whether the preceding soundscape can supply a causal prior. Our supervised data tuple and main model are

$$\begin{aligned} x_t &= N_t + s_t, \qquad \widehat{B}_t = P_\psi(H_t), \\ \widehat{R}_t &= \log|X_t| - \widehat{B}_t, \\ [X_t, \widehat{B}_t, \widehat{R}_t] &\xrightarrow{\text{Residual+BSR}} (\hat{s}_t, \hat{N}_t), \end{aligned} \tag{1}$$

where H_t excludes annotated call intervals and N_t is a real current no-call segment. Training has exact stems, while the separator never receives N_t or s_t as input. Here X_t denotes the current mixture STFT and $\widehat{B}_t$ its predicted background log-magnitude map. The 30-band FiLM precursor failed on held-out audio, motivating an information ladder and the 128-group residual interface in Fig. 1.

Time-domain separators [2], [3], attention models [4], [5], and band-split systems [6], [7] provide strong mixture-only backbones. Hybrid Transformers and MixIT broaden the architecture and supervision regimes [8], [9], but do not test a forecast of the upcoming background. FiLM, cross-attention, and cross-modal separation [10]–[12] motivate three interfaces. Forecasting baselines range from recurrent to linear and patch-based models [13]–[16].

Our contributions are: (1) a 128-group time-resolved causal background map with an explicit innovation and residual BSR interface;[1] (2) an annotation-aware construction of real clean ocean backgrounds and event-masked 60-s histories; (3) a six-way matched test separating adapter capacity, wrong-day background, persistence, learned forecast, and current Oracle information; and (4) a diagnostic chain from failed 30-band FiLM to the frequency-resolution ladder. Broad source and real-event studies assess coverage and transfer; they are secondary to the full-reference DCLDE test.

II. Method

A. High-resolution causal residual separator

The main history $H_t \in \mathbb{R}^{12\times 79\times 128}$ contains twelve strictly prior 5-s, 128-group log-magnitude maps. A 512-point Hann STFT with 128-point hop gives 79 frames per window at 2 kHz. Annotated event windows are masked and filled only with nearby clean windows within the past 60 s. A small residual GRU compresses each map into four temporal chunks, processes the twelve steps, and predicts the next full 79×128 map $\widehat{B}_t$; its zero-initialized decoder starts at last-map persistence. The forecaster is trained on clean training days, selected by validation map MAE, and frozen before separation.

We form $\widehat{R}_t = \log|X_t| - \widehat{B}_t$ on the same grid. The time-aligned pair $[\widehat{B}_t, \widehat{R}_t]$ is projected into mixture-derived BSR band tokens before axial attention. The residual projection starts at zero, so every conditioned separator initially equals the same transferred mixture-only BSR waveform mapping. We train with exact synthetic event stems and a waveform plus multi-resolution STFT loss. Figure 1 shows the data and inference paths.

B. Preliminary 30-band FiLM design

The precursor history $H_t \in \mathbb{R}^{12\times 30}$ contains 30-band log-power features computed on twelve preceding 5-s windows;

[1]Code and reproduction instructions: https://github.com/D-eval/ocean_noise_pred_splite.

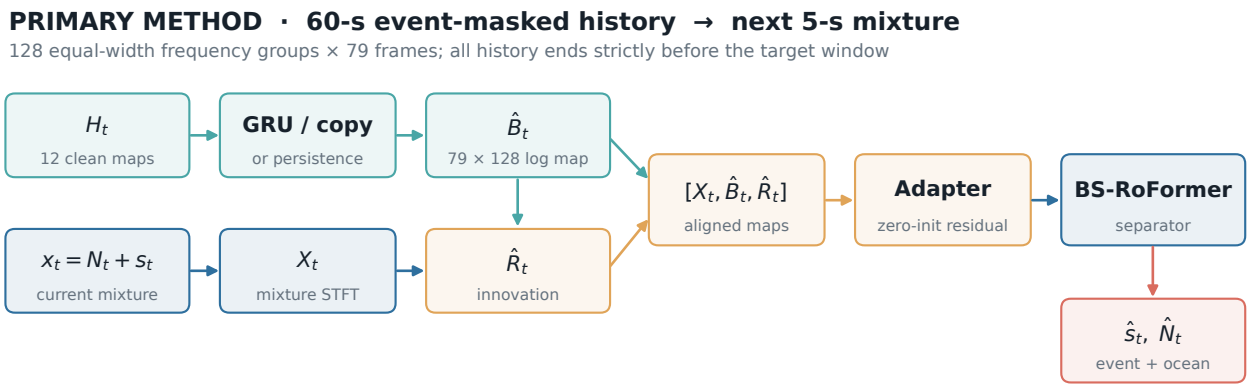


Fig. 1. Main 128-group causal residual architecture. The event-masked 60-s history predicts a full future 5-s background map; the explicit innovation is $\widehat{R}_t = \log|X_t| - \widehat{B}_t$. The same residual adapter is used for Zero, wrong-day Shuffled, Persistence, Forecast, and current-background Oracle conditions. Only the Oracle sees current clean background as an input.

annotated calls and their 2-s guards are excluded. Features are normalized using training data only. A residual GRU predicts

$$z_t = \mathrm{LN}(\mathrm{GRU}(H_t)), \quad \widehat{S}_t = S_{t-1} + g_\psi(z_t). \tag{2}$$

Residual initialization represents persistence exactly; a learned predictor is retained only if validation MAE improves upon persistence [17].

The precursor FC-BSR maps the mixture STFT to band tokens X_k and applies FiLM before the axial Transformer:

$$\widetilde{X}_k = (1 + \tanh\gamma_k) \odot X_k + \beta_k. \tag{3}$$

The separator predicts an event stem; the ocean stem is the mixture residual. Training combines waveform L_1 and multi-resolution complex-STFT losses. Concatenation and cross-attention were also tested as preliminary interfaces.

The preliminary 30-band audit holds the separator backbone, mixtures, initialization, and training rule fixed. A true BS-RoFormer bypasses the condition injector; Zero-FiLM retains the FiLM branch and the same frozen GRU, but replaces its output by $c = 0$. Shuffled-FiLM uses $E(N_j)$ from a deterministic half-cycle offset in the clean-window index of the same split ($j \neq t$); on the held-out test, this selects a different day. Persistence-FiLM uses S_{t-1}, and Forecast-FiLM uses the frozen GRU output. Oracle-FiLM receives $S_t^{\mathrm{oracle}} = E(N_t)$ computed from the actual current clean background after any shared mixture scaling. We call this the *30-band Oracle-FiLM*: E removes temporal structure and phase, so this row bounds neither a full-background condition nor the value of perfect forecasting with a richer representation. It is available only in the controlled experiment.

C. Marine-aware source distribution

The broad generator contains 13 families: harmonic, linear chirp, inharmonic, impulse train, frequency-modulated tone, band noise, whale upsweep, downsweep, arch, S-curve, pulse packet, vessel tonal, and vessel cavitation. Whale proxies use one to three localized events with variable duration, curvature, partials, taper, and flutter. Vessel-tonal and narrow-band cavitation components draw a 5–20 Hz low-frequency-oscillator amplitude envelope with probability 0.8; their broadband component is not modulated. Training SNR is uniform on $[-18, 12]$ dB. This distribution is intended to cover contour and modulation patterns, not to claim species identity.

We quantify spectral displacement with directional $D_{\mathrm{KL}}(p\|q)$ and finite symmetric Jensen–Shannon (JS) divergence [18]; spectral entropy and source–ocean overlap are reported separately. Larger KL/JS expands coverage but is not treated as semantic diversity. The DCLDE main experiment samples only its five whale and two vessel families; all six main-domain models see the same target distribution.

D. Main data construction and secondary ablations

The main dataset uses day-disjoint 2-kHz DCLDE 2013 recordings and whale-call annotations [19], [20]. Across 168 h of WAV, excluding annotated calls and a 2-s guard on each side yields 146.4 h of annotation-clean 5-s windows. For each target, the 12 immediately preceding 5-s blocks from the same WAV form a high-resolution history. Annotated calls and their 2-s guards are masked; 235 of 68,075 eligible current windows have one masked historical block, filled by the nearest clean block still within the preceding 60 s. Histories never cross WAV boundaries or use the current window. This leaves 94.5 h eligible overall, split into 37.3/9.7/47.6 h for train/validation/test. For each current clean N_t, a seven-family whale/vessel synthetic event s_t is mixed to form $x_t = N_t + s_t$. The main split is by recording day, and the GRU receives only event-masked features from the immediately preceding 5-s windows.

TABLE I
CONDITIONING-MODE ABLATION ON TOL-DERIVED SPECTRAL SURROGATES (DB).

Injection	Val. SI-SDRi	Test SI-SDRi	Paired Δ_0
None	4.864	4.059	0
Concatenation	5.343	4.432	+0.623
FiLM	**5.419**	**4.468**	**+0.692**
Cross-attention	5.128	4.261	+0.331

Secondary ablations use NOAA–Navy SanctSound hourly TOL products [21]. MB03 deployments 01–02/03/04 are train/validation/test. Random-phase inversion creates controlled spectral surrogates, not recovered audio. Concatenation, cross-attention, KL/JS coverage, and the one-hour history-resolution comparison are confined to this domain. The separate 42-h SanctSound WAV strength sweep also lacks event annotations, so it is a useful controlled real-background ablation but is not counted as clean-background primary evidence.

Complete-reference tiers report SI-SDR and SI-SDRi using scale-invariant projection [22], [23]. All uncertainty intervals are seeded paired bootstraps, blocked by recording day for the main test. The event classifier is frozen before any real-event evaluation. Current test features are used only as targets for the synthetic full-reference task, never as GRU input; all histories end before t.

III. EXPERIMENTS AND RESULTS

A. Preliminary conditioning-mode ablation

The initial 30-band design selected FiLM. Table I compares it with alternative interfaces on controlled 5-s spectral-surrogate examples; on 832 fixed-0-dB pairs (64 per family), FiLM improves 4.850 to 5.541 dB SI-SDRi. Band-wise scale and bias directly modulate existing band evidence. Concatenation must compress doubled features, while cross-attention must learn alignment from coarse predicted spectra under the same budget.

FiLM gains 0.869, 0.835, 0.919, 0.906, and 0.806 dB on the five whale proxy families. Vessel tonal is the hardest family: the conditioned absolute SI-SDRi is -8.295 dB, despite a 0.320-dB relative gain. Overall gain has weak correlation with JS ($r = -.057$) and entropy ($r = .122$), so increasing divergence broadens support but does not predict usefulness.

B. Preliminary 30-band FiLM audit on DCLDE

The original FC-BSR receives twelve event-masked preceding 5-s features and a current clean 5-s DCLDE segment; the seven-family event is added only after the clean background is selected. The model therefore sees x_t and H_t, while the exact N_t and s_t are supervision. The training split contains 37.3 h of eligible clean current audio. The 12-step GRU forecasts the next 5-s band spectrum; validation MAE is 1.926 dB versus 2.111 dB for persistence (8.8% lower). The older two-row comparison labeled a Zero-FiLM checkpoint as "BS-RoFormer." We retain its exact trained waveform mapping as the Zero control, attach the same frozen GRU in a bypassed branch, and train a true injector-bypassed BS-RoFormer. All six rows in Table II use the same mixtures, transferred backbone initialization, optimizer budget, validation selection, and 4,096 test examples from three held-out days. Figure 3 stratifies these same samples by input SNR.

The architecture effect, Zero minus BS-RoFormer, is -0.0002 dB (95% CI $[-0.0068, 0.0088]$). Forecast minus BS-RoFormer is -0.134 dB ($[-0.327, 0.125]$); Oracle minus BS-RoFormer is -0.096 dB ($[-0.238, 0.034]$). Oracle minus Forecast is only $+0.039$ dB ($[-0.090, 0.155]$). Shuffling and persistence also fail to show a positive average over Zero. Thus the matched controls do not establish that temporal forecast information improves this separator. Oracle's higher validation score (11.372 versus 10.946 dB for BS-RoFormer) does not transfer to a clear held-out gain. With only three test days, the intervals leave small effects unresolved; the 30-band oracle supplies more information, but separately optimized FiLM models do not form a guaranteed performance ordering.

a) Full-waveform oracle copy check.: To test whether losing information in E explains the negative 30-band result, we freeze the same trained BS-RoFormer and fit a pointwise, three-coefficient head on 8,192 training mixtures:

$$\hat{s}_t = w_b \hat{s}_{\mathrm{BSR}}(x_t) + w_x x_t + w_c c_t. \tag{4}$$

The head starts at $(1, 0, 0)$, exactly reproducing BSR, and its waveform-MSE global optimum is solved by least squares. We fit the *same* head with zero, wrong-time, and actual current background waveforms; validation and testing use 4,096 examples each, with the test IDs, SNRs, mixture scores, and frozen BSR outputs checked against Table II. For $c_t = N_t$, the learned weights are $(0, 1, -1)$ to numerical precision: the head copies the exact answer $x_t - N_t = s_t$. The maximum test waveform error is 5.96×10^{-8}; the finite SI-SDRi of 80.755 dB is limited by the metric's numerical floor. Zero and wrong-time heads receive no useful background identity and do not improve SI-SDRi (Table III). This is an intentionally noncausal information/path sanity check with a different interface, not an architecture-matched FiLM ablation or evidence that the 30-band forecast can recover a phase-aligned background waveform.

b) Oracle information ladder.: We next reduce the information in the *current* N_t while keeping the frozen BSR, the three-coefficient head, training examples, and 4,096 matched test mixtures fixed. A 512-point Hann STFT at 2 kHz has 257 frequency bins on a 64-ms hop. We average background power across either adjacent equal-width groups (128, 64, 32) or geometric 20–950-Hz groups (128, 64, 48, 30), and separately pool frames over 0.25, 1, or 5 s. The retained magnitude is inverted with either the actual N_t phase or the mixture x_t phase. Geometric groups absorb edge bins; only 95 of the nominal 128 geometric groups contain an STFT bin. The 30-band/5-s rung is analogous to, but not numerically identical with, the original $E(N_t)$.

Figure 2 identifies where this *copy path* ceases to show a held-out gain. Exact amplitude with mixture phase still adds 11.164 dB over BSR (day-block CI $[10.889, 11.594]$). With

TABLE II

PRELIMINARY 30-BAND FILM AUDIT ON 4,096 MATCHED ANNOTATION-CLEAN 5-S BACKGROUNDS PLUS SYNTHETIC EVENTS. VALUES ARE TARGET SI-SDRI (DB); BRACKETS ARE PAIRED 95% BOOTSTRAP INTERVALS OVER HELD-OUT RECORDING DAYS. THE 30-BAND ORACLE ALONE SEES $E(N_t)$.

Model	Validation	Test	Δ vs. BSR [95% CI]	Δ vs. Zero [95% CI]
BS-RoFormer	10.946	9.631	0	$+0.0002$ $[-0.009, 0.007]$
Zero-condition FiLM	10.967	9.631	-0.0002 $[-0.007, 0.009]$	0
Shuffled-condition FiLM	10.609	9.526	-0.105 $[-0.247, 0.020]$	-0.105 $[-0.256, 0.026]$
Persistence-condition FiLM	11.205	9.524	-0.107 $[-0.239, 0.083]$	-0.107 $[-0.232, 0.084]$
GRU-Forecast FiLM (FC-BSR)	11.213	9.497	-0.134 $[-0.327, 0.125]$	-0.134 $[-0.320, 0.125]$
30-band Oracle-FiLM, $E(N_t)$	11.372	9.535	-0.096 $[-0.238, 0.034]$	-0.095 $[-0.231, 0.035]$

TABLE III

FULL-WAVEFORM ORACLE COPY CHECK ON THE SAME 4,096 DCLDE TEST MIXTURES. TARGET SI-SDRI AND PAIRED DAY-BLOCK Δ AGAINST THE FROZEN BSR (DB).

Frozen BSR + fitted head	SI-SDRi	Δ [95% CI]
BSR; initial $(1, 0, 0)$	9.631	0
Zero waveform	9.231	-0.400 $[-0.483, -0.269]$
Wrong-time waveform	9.231	-0.400 $[-0.483, -0.269]$
Actual N_t waveform	80.755	$+71.124$ $[69.945, 72.081]$
Analytic $x_t - N_t$	80.755	$+71.124$ $[69.945, 72.081]$

equal-width groups, 128 retains $+2.067$ dB ($[0.969, 3.386]$), while 64 has -0.623 dB ($[-1.404, 0.259]$): the confirmed gain disappears between these levels. For geometric bands, nominal 128 already gives -0.529 dB ($[-1.257, 0.319]$), and 30 per-frame bands give -0.672 dB ($[-1.323, 0.086]$). Retaining the true phase at 30 per-frame bands gives only $+0.117$ dB ($[-0.671, 1.084]$). By contrast, keeping all 257 bins but pooling power to 1 s still yields $+0.585$ dB with mixture phase ($[0.285, 1.132]$); the 5-s version yields $+0.249$ dB ($[-0.044, 0.495]$). Thus this interface is especially sensitive to frequency averaging and the grouping rule, while time averaging alone is less destructive. These are sufficiency results for the fixed waveform copy head, not information-theoretic limits or new FiLM results.

C. Main 128-group residual experiment

a) Trained high-resolution Oracle residual separator.: We next test the representation in an actual BS-RoFormer, rather than only a frozen linear copy head. The actual current N_t and mixture x_t are transformed with a 512-point, 128-hop STFT. Mean power in 128 equal-width frequency groups yields time-resolved $\log |B_t|$; the explicit innovation is $R_t = \log |X_t| - \log |B_t|$ on the same grid. These maps are interpolated to the BSR frame grid and projected into its mixture-derived band tokens before axial attention. This zero-initialized residualized $[X_t, B_t, R_t]$ input uses neither FiLM nor a forecaster; its initial waveform output exactly matches ordinary BSR (maximum difference zero).

Both models use the same day-disjoint DCLDE mixtures, transferred backbone initialization, seed, batch order, 6,000 updates, AdamW/cosine schedule, loss, and validation-best checkpoint rule. The existing BSR selected step 5,300; the Oracle model selected step 6,000. On 4,096 matched test examples, BSR scores 9.631 and Oracle-Residual-BSR 16.767 dB target SI-SDRi. The paired improvement is $+7.136$ dB (recording-day 95% interval $[6.970, 7.282]$), with all three held-out days positive. The combined high-resolution prior and explicit innovation interface therefore benefits this controlled separator. Actual N_t is unavailable at deployment; this two-model comparison also does not isolate adapter capacity from the condition information.

b) Causal high-resolution residual conditions.: We then replace the current-background Oracle map with two conditions computed strictly from the preceding 60 s. Each of twelve 5-s history windows has the same 79×128 log-magnitude grid. Annotation-overlapping windows are masked; when one is unavailable, its slot copies the nearest annotation-clean window still inside the past 60 s. This affects 235 of 68,075 eligible current windows. Persistence uses the last clean history map. A lightweight GRU encodes each map into four temporal chunks, processes twelve steps, and decodes a residual correction to persistence. It predicts the entire future 79×128 log-background map, without seeing the current waveform. The predictor is trained for 3,000 updates on training-day clean backgrounds and frozen at its lowest validation log-magnitude MAE. Validation MAE is 0.5456 for the GRU versus 0.5518 for persistence (1.12% relative gain). On the held-out test days, the corresponding map MAEs are 0.5308 and 0.5386.

For every condition map B, we form $R = \log |X_t| - B$ and feed $[X_t, B, R]$ through precisely the Oracle residual adapter. The Zero control repeats the fixed training-set mean log background at every frame. The Shuffled control draws an annotation-clean B_j from a *different recording day*: training and testing draw within their respective splits; validation draws from training days because validation has only one day. Persistence takes the last clean history map; Forecast uses the frozen GRU. All four separators start from the same transferred BSR backbone and zero adapter, so their initial waveform output is identical to ordinary BSR. They use identical synthetic events, training-example order, 6,000 update budget, optimizer, loss, validation-best checkpoint rule, and held-out mixtures. Thus the four residual arms differ only in the source of B.

Table IV separates interface and background effects. Zero exceeds ordinary BSR by $+2.525$ dB (recording-day paired 95% CI $[2.262, 2.701]$); Shuffled exceeds BSR by $+2.313$ dB ($[2.027, 2.905]$). Hence the full BSR-to-causal gain cannot be attributed to correct history alone. The Zero contrast

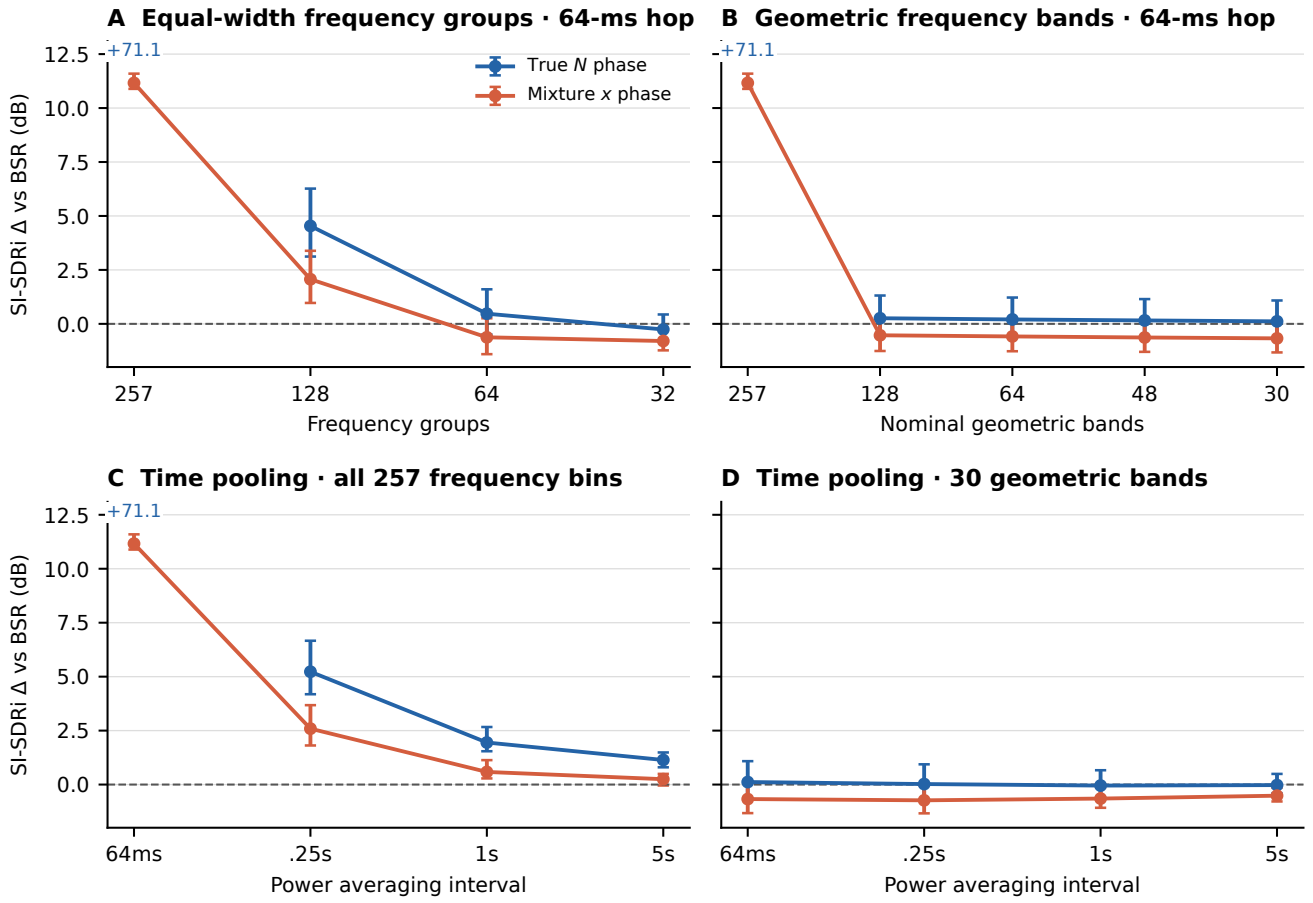


Fig. 2. Current-background Oracle information ladder on identical DCLDE test mixtures. Each point fits the same three-coefficient copy head on training mixtures. Vertical bars are paired 95% day-block intervals for SI-SDRi difference from frozen BSR. Full-resolution true-phase N_t gives +71.1 dB and is above the plotted range; mixture phase is a separate information loss. The threshold depends on frequency grouping and this copy interface.

TABLE IV
MATCHED DCLDE TEST ON 4,096 CLEAN-BACKGROUND MIXTURES. ALL FIVE RESIDUAL ARMS USE THE IDENTICAL 128-GROUP ADAPTER; ZERO USES A FIXED TRAINING MEAN MAP AND SHUFFLED USES ANOTHER RECORDING DAY. ONLY ORACLE RECEIVES THE CURRENT TRUE BACKGROUND. VALUES ARE MEAN TARGET SI-SDRI IN DB.

Condition	SI-SDRi	Gain vs. BSR
BSR (mixture only)	9.631	—
Zero-Residual	12.156	+2.525
Shuffled-Residual	11.944	+2.313
Persistence-Residual	14.295	+4.664
GRU-Forecast-Residual	14.351	+4.720
Oracle-Residual	16.767	+7.136

includes both the extra adapter and the current mixture log-spectrum supplied by the innovation path, so it does not separately identify parameter count. With the adapter held fixed, Persistence exceeds Zero by $+2.138$ dB ($[1.775, 2.377]$) and wrong-day Shuffled by $+2.351$ dB ($[1.571, 2.661]$), positive on every held-out day. This supports a contribution from temporally relevant historical background beyond architecture and arbitrary background maps. Forecast exceeds Persistence by only $+0.056$ dB ($[0.036, 0.074]$), again positive on all three days but small in practice. Forecast retains 66.1% of the Oracle gain over BSR, with a 2.416 dB Oracle gap ($[2.114, 2.640]$).

D. Secondary ablations: time scale and target strength

The time-resolution diagnostic in Table V is an ablation on continuous SanctSound audio, not the main annotation-clean data. Only its 5-s predictor beats persistence on validation (MAE 1.366 versus 1.386 dB); the other scales select persistence. The separate 42-h strength curve also uses unlabeled current background. It shows that forecast conditioning is strength-dependent in that domain and must not be read as a result on annotation-clean current segments.

The harder fixed-5-s experiment on unannotated SanctSound WAV compares Zero-condition FiLM, a current-spectrum oracle, causal forecast, and a time-shuffled condition on 1,152 paired test mixtures. This oracle belongs to the secondary data domain; Table II uses the actual annotation-clean DCLDE N_t.

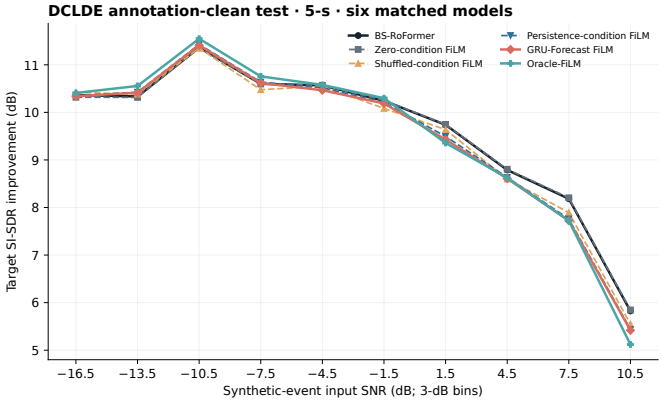


Fig. 3. Preliminary 30-band FiLM audit on annotation-clean DCLDE backgrounds: mean target SI-SDRi in 3-dB input-SNR bins for all six matched methods. Table II gives day-block uncertainty for their overall means.

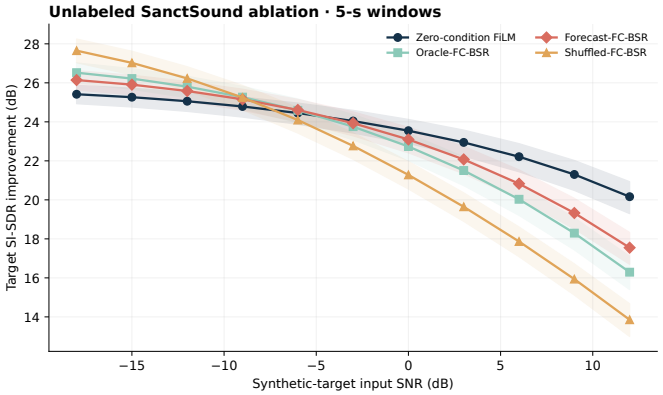


Fig. 4. Ablation on 5-s unlabeled SanctSound background: target-strength sweep. Lines show the four conditions; shaded bands are paired 95% intervals.

TABLE V
ABLATION: PREDICTOR-RESOLUTION DIAGNOSTIC ON SANCTSOUND BACKGROUND.

Step	Predictor	Cond.	Base	Gain [95% CI]
5 s	GRU	29.753	28.527	1.226 [1.082,1.381]
30 s	Persist.	29.630	28.534	1.095 [0.935,1.265]
1 min	Persist.	29.661	28.533	1.128 [0.971,1.298]
10 min	Persist.	29.711	28.535	1.176 [1.036,1.326]

Cond./Base/Gain are target SI-SDRi in dB.

Averaged across training SNRs, the baseline scores 23.470 dB and Forecast-FC-BSR 22.871 dB ($\Delta = -0.599$, CI $[-0.851, -0.346]$); even the oracle is lower. This exposes conditioning/capacity effects that the simpler diagnostic misses. Figure 4 gives the more informative strength sweep. Forecast conditioning helps weak targets by 0.728, 0.643, 0.525, 0.366, and 0.157 dB at $-18, -15, -12, -9, -6$ dB, respectively, then loses from -3 dB onward. The shuffled condition also helps the weakest targets but collapses as SNR rises; therefore weak-target gain alone cannot prove that temporal forecast accuracy, rather than regularization or extra capacity, is responsible.

E. Real events without separated ground truth

Following learned reference-free metrics [24], [25] and semantic audio embeddings [26], [27], we train, calibrate, and freeze a binary event classifier (test AUROC .9723, AP .9755, F1 .9275). Mixture consistency imposes $\hat{N} = x - \hat{s}$. We restore $\hat{s}$ at gains $\{-60, -30, -24, -18, -12, -6, 0\}$ dB and test whether event probability recovers monotonically. Identity and all-to-event controls reject trivial solutions; changes outside annotated intervals audit preservation.

We evaluate the four frozen 128-group residual models on the exact 103 test calls used by the preliminary FiLM audit. Each call receives twelve strictly earlier annotation-clean maps from the same WAV. Six histories each lack one clean window, filled by the nearest clean window still inside the prior 60 s. Zero uses the training mean, Shuffled a clean background from another test day, Persistence the last historical map, and Forecast the frozen GRU applied to that history. No condition reads the current call. Table VI and Fig. 6 show the results.

Persistence versus Zero lowers $p(\hat{N})$ by .0624 (recording-day paired 95% CI $[-.0682, -.0547]$), raises $p(\hat{s})$ by .0469 ($[.0198, .1006]$), and lowers off-event change by .0255 ($[-.0386, -.0077]$). Forecast shows the same directions versus Zero, with $\Delta p(\hat{N}) = -.0531$ and Δ off-event $= -.0260$. However, wrong-day Shuffled attains lower $p(\hat{N})$ and higher $p(\hat{s})$ than either causal condition: Persistence minus Shuffled is $+.0471$ ($[.0317, .0706]$) for $p(\hat{N})$ and $-.0606$ ($[-.0731, -.0404]$) for $p(\hat{s})$. Persistence protects off-event audio much better ($\Delta = -.0408$, $[-.0519, -.0320]$). Monotonicity is near saturation for all four models and does not separate them. The classifier probabilities therefore do not reproduce the controlled full-reference ranking; they can reward aggressive extraction, while off-event preservation favors the causal conditions. Isolated real-event stems remain unavailable.

F. Classifier floor and synthetic-target coverage

To audit the high $p(\hat{N})$, we apply the frozen classifier to annotation-clean test audio, truncated to the separator's 9,984 output samples. Its mean probability is .153 on 4,096 day-proportional clean windows and .176 on 103 clean windows from the same WAV, immediately before the corresponding real calls; both medians are .093. The real mixtures average .909. The four separated-background means of .781–.891 are well above the clean floor; paired differences from the same-WAV control range from .606 to .715. This is a classifier-response gap, not a waveform measure of leaked whale energy: artifacts or unmarked events can also elicit responses.

We generate 32 whale-target mixtures per matched clean window (3,296 total), using the five training whale families and $[-18, 12]$ dB sampled strengths. Their mean classifier probability is .407. For a proxy coverage test, each event-bearing clip's frozen 64-D penultimate activation is subtracted from its same-WAV clean activation. After scaling dimensions by synthetic variability, median real-to-real leave-one-out nearest-neighbor distance is .233. Searching all 3,296 synthetic clips gives a real-to-synthetic median of .262, but this reference

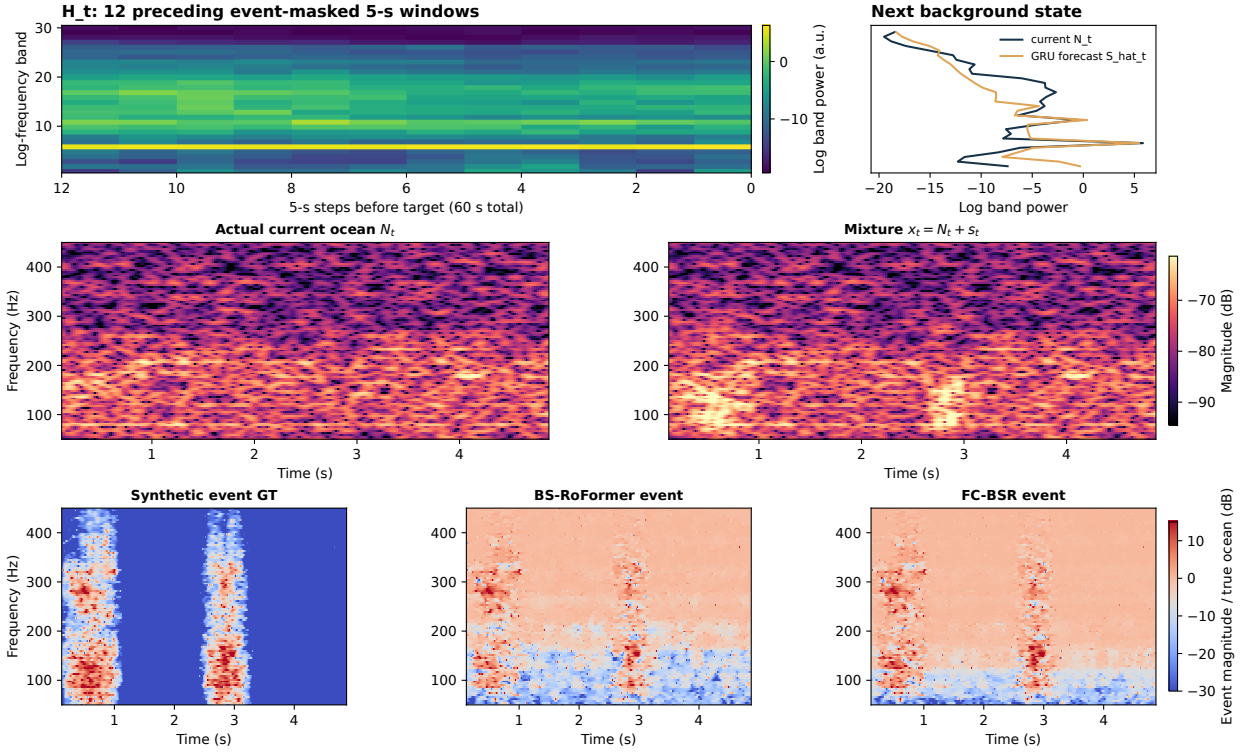


Fig. 5. Preliminary 30-band FiLM illustration: median-selected -12 dB synthetic whale pulse packet in an annotation-clean 5-s DCLDE recording. The ocean and mixture share an absolute magnitude scale; the lower row shows each event stem relative to the actual ocean magnitude on a common dB scale. This example does not show the new 128-group residual model. Neither estimate fully removes background texture; aggregate preliminary results are in Fig. 3.

TABLE VI
FROZEN 128-GROUP RESIDUAL MODELS ON THE SAME 103 REAL CALLS. ALL MODELS SEE THE SAME MIXTURE; REAL SEPARATED STEMS ARE UNAVAILABLE. OFF-EVENT IS RELATIVE BACKGROUND CHANGE OUTSIDE ANNOTATED CALLS.

Method	$p(x)$	$p(\hat{N}) \downarrow$	$p(\hat{s}) \uparrow$	Mono.↑	Off-event↓
Zero-Residual	.909	.891	.802	.990	.062
Shuffled-Residual	.909	.781	.910	.989	.078
Persistence-Residual	.909	.829	.849	.992	.037
GRU-Forecast-Residual	.909	.838	.842	.989	.036

set is much larger. Using one synthetic clip per real call gives 103 candidates per repeat; across 32 repeats the median distance is .483, or 2.08× the real-to-real distance. A median 51.5% of calls exceed the real-to-real 95th-percentile distance threshold (versus 3.9% with all 3,296 synthetic candidates). Synthetic examples are thus sparse near real events in this *classifier-dependent, equal-size* representation. This motivates target-distribution refinement; neither the embedding nor detector confidence establishes the physical cause of residual event responses without isolated real whale stems. Real background evolution since the preceding clean window also enters this embedding difference.

IV. DISCUSSION AND CONCLUSION

The main-domain 30-band Oracle is a diagnostic of the existing interface: even when FiLM receives $E(N_t)$ from the exact current clean waveform, its held-out mean does not exceed the mixture-only BS-RoFormer, and its paired interval spans zero. The full-waveform copy check succeeds exactly, so the current negative result cannot be interpreted as evidence that the true background lacks useful information. It leaves the relative roles of 30-band compression and FiLM injection unresolved. The fixed-head ladder shows a clear representation bottleneck: even mixture-phase conditions retain a confirmed gain at 128 equal-width frequency groups or 1-s full-bin temporal pooling, whereas 64 equal-width groups

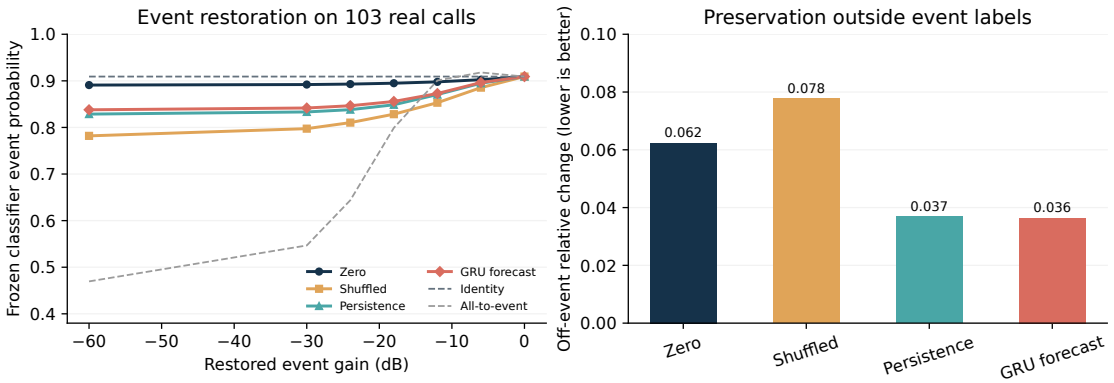


Fig. 6. Frozen-classifier event restoration and off-event preservation for four matched residual models. Identity and all-to-event are degenerate controls. Shuffled scores strongly on event probabilities but changes more audio outside annotated calls than either causal condition.

and 30 geometric bands do not. The original FiLM result cannot be assigned to this bottleneck alone because its interface differs. The trained 128-group Oracle-Residual-BSR gains $+7.136$ dB on matched held-out mixtures, showing that a high-resolution true background and explicit innovation can help the BSR backbone. Replacing true current background with past-60-s persistence or a frozen GRU forecast retains large gains of $+4.664$ and $+4.720$ dB over BSR. The fixed-mean Zero and wrong-day Shuffled controls also gain $+2.525$ and $+2.313$ dB, respectively; therefore part of the BSR-to-causal gain comes from the expanded interface. Holding that interface fixed, Persistence gains $+2.138$ dB over Zero and $+2.351$ dB over Shuffled. Correctly timed history therefore provides useful information beyond a generic background map. The forecast adds a consistent but modest $+0.056$ dB over persistence on three test days; the remaining 2.416 dB Oracle gap shows headroom for a better causal prior. The Zero control cannot distinguish parameter capacity from access to the mixture log spectrum in the innovation path. Synthetic-target supervision and only three held-out days also limit generalization. Zero-FiLM is effectively tied with BS-RoFormer, while Forecast is no better than either Zero or Shuffled. The result does not isolate prediction error as the bottleneck; a five-second, 30-band static summary and its FiLM interface may be too coarse for the separator's time-frequency tokens. FiLM remains the best interface on controlled spectral surrogates, and the secondary unlabeled WAV sweep shows weak-target gains, but neither establishes a main-domain gain. The new real-call audit has a mixed outcome: causal maps improve all three semantic/preservation measures against Zero, but Shuffled wins the two classifier event probabilities while losing markedly on off-event change. Forecast is not uniformly better than Persistence. Thus the real calls do not establish that correctly timed history improves semantic separation; they expose a preservation-versus-detection tradeoff under a surrogate classifier. The old 30-band FiLM real-call comparison remains preliminary. Only three held-out days and no isolated real stems limit this inference. The clean-control audit rules out a uniformly high classifier floor as the sole explanation of $p(\hat{N})$, while the density-matched embedding audit finds a synthetic-to-real target coverage gap in classifier space. Better target sampling is a testable next step; the current evidence does not isolate this gap from separation artifacts or annotation uncertainty.

The existing 30-band FiLM frontend is usable for inspection: on a CPU smoke test, one 5-s chunk ran at RTF 0.014 after history features were prepared; the interface displays forecasts and permits playback and download of both estimates. This does not establish deployment of the new 128-group residual model or field use. Further work should improve prediction of event-relevant time-frequency structure rather than only mean log-magnitude MAE, cover more sites and seasons, and obtain isolated or stronger partial references for real events. Classifier-based restoration is evidence of semantic transfer, not a replacement for waveform fidelity.

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